A parabolic subgroup and a conic in the rank-five construction

Human Compression project

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We give the subgroup and intersection arguments omitted from Section 4 of Leemans–Vandenschrick [1]. The four source matrices are unchanged. The two signs require different geometry: the second end fixes a point for one sign and preserves a conic for the other. We retain the classical rotation correspondence, its recursive intersection criterion [1, Lemmas 2.2 and 2.3], and the description of the automorphisms of $\mathrm{PSL}_3(F)$ as semilinear collineations and their compositions with inverse transpose. No classification of subgroups of $\mathrm{PSL}_3(F)$ or $\mathrm{PGL}_2(F)$ enters the proof.

Throughout, $p \geq 5$ is prime, w has order three, and $F = \mathbb{F}_p(w)$. Thus $F = \mathbb{F}_p$ when $p \equiv 1 \pmod 3$, and $F = \mathbb{F}_{p^2}$ when $p \equiv -1 \pmod 3$. Put $q = |F|$ and $k \in \{1, -1\}$, and set

$$a = \mathrm{diag}(1, w, w^2), \quad b = \begin{pmatrix} -1 & 0 & -k \\ 0 & -w^2 & -kw^2 \\ 0 & 0 & w \end{pmatrix}, \quad c = \begin{pmatrix} 1 & k & 0 \\ 0 & k & 1 \\ 0 & -1 & 0 \end{pmatrix}, \quad d = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

These matrices have determinant one. When discussing projective groups we use the same letters for their images. Write $e_1 = [1, 0, 0]^t$ and $\mu_6 = \langle -w^2 \rangle \leq F^\times$.

Theorem 1. *The marking (a, b, c, d) generates $\mathrm{PSL}_3(F)$, satisfies the string rotation relations and the full rotation intersection property, and has type $[3, 6, 6, 3]$ when $k = 1$ and $[3, 6, 3, 3]$ when $k = -1$. If $F = \mathbb{F}_p$, it defines a chiral polytope. If $F = \mathbb{F}_{p^2}$ with $p \equiv -1 \pmod 3$, it defines a directly regular polytope with rotation group $\mathrm{PSL}_3(F)$; its full automorphism group is the extension by the Frobenius automorphism, of order twice that of the rotation group.*

1 One projection and its translation kernel

We first work in $\mathrm{SL}_3(F)$. Let $\widetilde{A} = \langle a, b, c \rangle$. Every element has block form

$$\begin{pmatrix} \epsilon & v \\ 0 & M \end{pmatrix}, \qquad \epsilon \in \{1, -1\}, \quad \det M = \epsilon.$$

Projection to M will be denoted by π. Put $t = k(2w + 1)$; it is nonzero because $(2w + 1)^2 = -3$. Direct multiplication gives

$$u = b^2 a^{-1} = I + tE_{13} - tE_{23}.$$

The lower block of u is an upper unipotent matrix with parameter $-t$. Conjugation by the lower block of a multiplies that parameter by w^2. Since $F = \mathbb{F}_p(w)$, additive combinations give every upper unipotent matrix in $\mathrm{SL}_2(F)$. The lower block of c is

$$\begin{pmatrix} k & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & -k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Consequently the image contains the matrix interchanging the two root groups, and then all of $\mathrm{SL}_2(F)$ by elementary elimination. The lower block of b has determinant -1, so

$$\pi(\widetilde{A}) = \{M \in \mathrm{GL}_2(F) : \det M = \pm 1\}.$$

The kernel is an additive subgroup V of the row space F^2, normal under the natural action of $\mathrm{SL}_2(F)$. It cannot be zero. If it were, the unique lift of the central matrix $-I_2$ would be central in $\widetilde{A}$ and have the form

$$z = \begin{pmatrix} 1 & v \\ 0 & -I_2 \end{pmatrix}.$$

Commutation with a forces $v = 0$, since $w, w^2 \neq 1$. But $\mathrm{diag}(1, -1, -1)$ does not commute with c: its nonzero entry k in position $(1, 2)$ changes sign. This is impossible in odd characteristic. Thus $V \neq 0$. The action of $\mathrm{SL}_2(F)$ is transitive on nonzero row vectors, so normality forces $V = F^2$. We have proved

$$\widetilde{A} = \left\{ \begin{pmatrix} \epsilon & v \\ 0 & M \end{pmatrix} : \epsilon = \pm 1,\ \det M = \epsilon,\ v \in F^2 \right\}. \tag{1}$$

Its intersection with the scalar center of $\mathrm{SL}_3(F)$ is trivial. The projective image A therefore has order $2q^3(q^2 - 1)$. Equivalently, A consists of the projective transformations fixing e_1 whose eigenvalue there, in an $\mathrm{SL}_3(F)$ representative, belongs to μ_6. Multiplication by a central cube root changes that eigenvalue to 1 or -1. In particular A is generally smaller than the full point stabilizer.

Equation (1) supplies all $I + sE_{12}$ and $I + sE_{13}$. Conjugation by d cycles the coordinate indices and supplies the other four elementary root groups. Hence

$$\langle a, b, c, d \rangle = \mathrm{SL}_3(F)$$

before taking projective images.

2 The first adjacent pair

Let $T = \{I + sE_{13} + tE_{23} : s, t \in F\}$ and let $D_0 = \mathrm{diag}(w, w^2)$ act on its two parameters. The vector of $u = b^2a^{-1}$ is $t(1, -1)$. Its orbit spans exactly two dimensions over $\mathbb{F}_p$: the vectors $t(1, -1)$ and $t(w, -w^2)$ are independent, and $D_0^2 + D_0 + I = 0$. Denote this additive subgroup by V_0. Conjugation by b acts on the parameters as $-D_0^2$, so preserves V_0; conversely its square acts as D_0. Since $a = u^{-1}b^2$,

$$H = \langle a, b \rangle = V_0 \rtimes \langle b \rangle, \qquad |H| = 6p^2. \tag{2}$$

Here b has order six, and its nonidentity powers have nontrivial diagonal image. When $F = \mathbb{F}_p$, $V_0 = T$; when $F = \mathbb{F}_{p^2}$, it has order q, not q^2. All elements of H fix both e_1 and e_2.

For $1 \leq j \leq 5$ the entry $(b^j)_{13}$ is nonzero. Indeed it is

$$-k\,\frac{(-1)^j - w^j}{-1 - w},$$

and $-w^2$ has order six. Thus no such power is diagonal, proving $\langle a \rangle \cap \langle b \rangle = 1$ even projectively.

3 The point case

Suppose $k = 1$ and put $r = [1, w, w^2]^t$. Both b, c fix e_1 and r. The source matrix

$$J(w) = \begin{pmatrix} w & w^2 & 1 \\ w^2 & w & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

has determinant $-3(2w+1) \neq 0$. Literal multiplication gives

$$J^{-1}a(w)J = w^2d^{-1}, \qquad J^{-1}b(w)J = wc(w^2)^{-1},$$
$$J^{-1}c(w)J = w^2b(w^2)^{-1}, \qquad J^{-1}dJ = wa(w^2)^{-1}.$$

Thus it exchanges the marking with the inverse-reversed marking at w^2, projectively. Applying (1) at w^2 shows that $B = \langle b, c, d\rangle$ is the corresponding restricted stabilizer of r: its allowed eigenvalues at r are precisely μ_6.

Choose the basis (e_1, r, e_2), represented by

$$P = \begin{pmatrix} 1 & 1 & 0 \\ 0 & w & 1 \\ 0 & w^2 & 0 \end{pmatrix}.$$

In that basis the two middle generators are

$$b' = \operatorname{diag}(-1, w, -w^2), \qquad c' = \begin{pmatrix} 1 & 0 & -w^2 \\ 0 & -w^2 & -w \\ 0 & 0 & -w \end{pmatrix}.$$

Their commutator $b'c'b'^{-1}c'^{-1}$ has translation vector $(-(2w+1), w)$ in positions $(1,3),(2,3)$. Conjugation by b' acts on these two coordinates with multipliers w and $-w^2$.

The prime-field minimal polynomials of these multipliers are distinct and relatively prime. In the prime-field case this follows from their different values; in the quadratic case their roots have orders three and six, respectively. Polynomial projections in this action therefore separate the two coordinates. Each nonzero coordinate then generates all of F under its multiplier, since $F = \mathbb{F}_p(w)$. Consequently $E = \langle b, c\rangle$ contains the full two-dimensional translation group in this basis.

Modulo that group, the two eigenvalue pairs at (e_1, r) are $(-1, w)$ and $(1, -w^2)$. They generate all $\{\pm 1\} \times \mu_6$: the second pair has order six and cannot supply the first coordinate -1. They therefore give all twelve allowed diagonal classes. After choosing the unique scalar adjustment making the first eigenvalue ± 1, these are exactly the elements of $A \cap B$. We conclude that

$$A \cap B = E, \qquad |E| = 12q^2. \tag{3}$$

The action of V_0 on r, whose third coordinate is nonzero, is free. Thus the stabilizer of r in H has order at most six. It contains $\langle b\rangle$, so $H \cap E = \langle b\rangle$. Also $\langle b\rangle \cap \langle c\rangle = 1$: in the displayed basis b' is diagonal, whereas the $(1,3)$ entry of $(c')^j$ is nonzero for $1 \leq j \leq 5$, because its two diagonal values have ratio of order six. These statements, (2), and the rotation criterion prove the full intersection property for (a, b, c).

The same conclusion at w^2, transported by J, proves the intersection property for (b, c, d). In particular $E \cap \langle c, d\rangle = \langle c\rangle$. Combining this with (3) gives

$$A \cap \langle c, d\rangle = \langle c\rangle.$$

Finally $A \cap \langle d\rangle = 1$, since d cycles the three coordinate points and A fixes e_1.

4 The conic case

Suppose $k = -1$. The conic in the source is

$$xy + wyz + w^2zx = 0.$$

Set $U = x+wz$, $V = z$, $W = y+w^2z$. Its equation becomes $UW - V^2 = 0$, with parametrization $[s : t] \mapsto [s^2 : st : t^2]$. For $g = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$, define

$$\iota(g) = \frac{1}{\det g} \begin{pmatrix} \alpha^2 & 2\alpha\beta & \beta^2 \\ \alpha\gamma & \alpha\delta + \beta\gamma & \beta\delta \\ \gamma^2 & 2\gamma\delta & \delta^2 \end{pmatrix}.$$

This is the symmetric-square representation divided by its determinant; it is a homomorphism into $\mathrm{SL}_3(F)$ and is unchanged by scalar multiplication of g. Its projective action is faithful on $\mathrm{PGL}_2(F)$: an element fixing every quadratic point fixes every point of the projective line and is scalar.

Let T denote the above coordinate change. Exact multiplication gives

$$w^2 TbT^{-1} = \iota \begin{pmatrix} -w^2 & 0 \\ 0 & 1 \end{pmatrix}, \quad wTcT^{-1} = \iota \begin{pmatrix} w & -w^2 \\ 0 & 1 \end{pmatrix}, \quad w^2 TdT^{-1} = \iota \begin{pmatrix} -w^2 & 1 \\ -w & 0 \end{pmatrix}.$$

The scalar factors have order three and disappear projectively. We may therefore work with these three 2×2 matrices, denoted by $\bar{b}, \bar{c}, \bar{d}$.

The first two act on an affine coordinate as

$$z \mapsto -w^2 z, \qquad z \mapsto wz - w^2.$$

Their commutator is a nonzero translation. Conjugation by their multipliers gives its entire F-span over $\mathbb{F}_p$, hence all translations, while the multipliers generate μ_6. Thus

$$\langle \bar{b}, \bar{c} \rangle = \{ z \mapsto tz + h : t \in \mu_6,\ h \in F \}, \qquad |E| = 6q. \tag{4}$$

The element $\bar{d}$ sends infinity to w. Conjugating the full translation group and then translating w to zero gives the full opposite unipotent root group. The two root groups generate $\mathrm{PSL}_2(F)$. The determinants of $\bar{c}, \bar{d}$ are w, a square; the determinant of $\bar{b}$ is $-w^2$. It follows directly from the determinant-square quotient that

$$B \cong \begin{cases} \mathrm{PSL}_2(F), & -1 \text{ is a square in } F, \\ \mathrm{PGL}_2(F), & -1 \text{ is not a square in } F. \end{cases}$$

No subgroup classification is needed for this conclusion.

Under the coordinate change, e_1 is the conic point infinity and e_2 is the point zero. An element of B fixing e_1 is represented by $\begin{pmatrix} t & h \\ 0 & 1 \end{pmatrix}$. Its eigenvalue at e_1 in the displayed determinant-one lift is t. By (1), it belongs to A exactly when $t \in \mu_6$. These multipliers all occur in (4), so once again

$$A \cap B = E.$$

Since H also fixes e_2, its intersection with E fixes zero in the affine model. This subgroup is exactly $\langle \bar{b} \rangle$; therefore $H \cap E = \langle b \rangle$. The affine transformation $\bar{c}$ has order three, and neither of its nonidentity powers fixes zero. Hence $\langle b \rangle \cap \langle c \rangle = 1$. Together with the first cyclic intersection, this proves IP^+ for (a, b, c).

The matrices c, d permute the four points

$$e_1, e_2, e_3, [1, 1, 1]^t$$

as two three-cycles: c fixes e_1, and d cycles e_1, e_2, e_3. These points form a projective frame, so the action is faithful and $\langle c, d \rangle \cong A_4$. Its stabilizer of e_1 is $\langle c \rangle$, giving

$$A \cap \langle c, d \rangle = \langle c \rangle, \qquad A \cap \langle d \rangle = 1.$$

This supplies the actual conic intersection argument, rather than the point-pair shorthand printed at the end of source Section 4.

5 The full intersection property and orientation

For either sign we have proved IP^+ for (a, b, c) and all three recursive intersections

$$A \cap \langle b, c, d\rangle = \langle b, c\rangle, \quad A \cap \langle c, d\rangle = \langle c\rangle, \quad A \cap \langle d\rangle = 1.$$

The retained recursive criterion therefore gives the full rank-five rotation intersection property. The six interval products are nonidentity projective involutions. Literal multiplication gives squares I for $ab, bc, cd, abc, abcd$, and $(bcd)^2 = w^2 I$. Their nontriviality is apparent from the nonzero off-diagonal entries. The eigenvalues of a, b, c, d give their projective orders $3, 6, 6, 3$ for $k = 1$ and $3, 6, 3, 3$ for $k = -1$.

Suppose first that $F = \mathbb{F}_p$. An automorphism reversing the marking must invert a and fix the three prefix involutions $ab, abc, abcd$. It preserves $A = \langle a, ab, abc\rangle$. This group fixes e_1 and fixes no line: its translation subgroup rules out lines not containing e_1, and its lower $\mathrm{SL}_2(F)$ rules out a fixed line through e_1. An automorphism involving inverse transpose would send its fixed point to a fixed line, so cannot preserve A. There is no nontrivial field automorphism. The proposed reverser is therefore conjugation by a matrix $P \in \mathrm{GL}_3(F)$.

Projective commutation with a determinant-one involution whose square is I is ordinary commutation: its scalar multiplier has both square and cube one, so equals one. Here

$$ab = \begin{pmatrix} -1 & 0 & -k \\ 0 & -1 & -k \\ 0 & 0 & 1 \end{pmatrix}, \quad abc = -P_{23}, \qquad abcd = -P_{12},$$

where P_{ij} is the coordinate transposition matrix. Commuting with P_{12}, P_{23} forces $P = \alpha I + \beta \mathcal{J}$, with $\mathcal{J}$ the all-ones matrix. The $(3, 1)$ entry of $\mathcal{J}(ab) - (ab)\mathcal{J}$ is -2, so $\beta = 0$. Thus P is scalar and cannot invert the element a of order three. This proves chirality over the prime field for both signs.

If $F = \mathbb{F}_{p^2}$ with $p \equiv -1 \pmod 3$, Frobenius sends w to w^2 and hence a to a^{-1}. All three displayed prefix involutions are defined over $\mathbb{F}_p$, so Frobenius fixes them. It is an involutory automorphism of the rotation group and is outer. The retained correspondence gives the directly regular extension by this automorphism; its full group, rather than its rotation subgroup, has order $2|\mathrm{PSL}_3(F)|$.

6 The six marking classes and attribution

Over $\mathbb{F}_p$, the two choices w, w^2 give different ordered marking classes for each sign. Indeed they are exactly the enantiomorphic markings: $a(w^2) = a(w)^{-1}$, $b(w^2) = a(w)^2 b(w)$, and c, d are unchanged. An isomorphism of these ordered markings would be the reverser just excluded. For $k = 1$ the matrix J shows that their two dual markings give the same two classes. For $k = -1$ the dual type is $[3, 3, 6, 3]$, different from both original types; it contributes two further classes. Thus the source construction gives six ordered, oriented marking classes, including duals.

The word "polytopes" in the source's count must be read with this orientation convention. The two enantiomorphic markings are the two choices of base-flag orbit in one unbased chiral polytope. If orientation is forgotten, the above six classes give three unbased isomorphism classes: one of type $[3, 6, 6, 3]$, and a dual pair of the other two types. The first is self-dual after orientation is forgotten. These distinctions follow from the retained rotation correspondence; they do not alter the source's existence theorem.

The matrices, the two signs, the conic, the duality matrix, and the prime and quadratic-field constructions are the source authors' work. The projection-kernel argument, the separated translation modules, the explicit symmetric-square model and the full recursive intersections above provide the omitted uniform proofs. We also make the full regular group and the orientation convention explicit. This component treats Theorem 1.2 and Section 4 only. It does not settle the source's later classifications, global rank obstruction, or open questions.

References

[1] D. Leemans and A. Vandenschrick, *On chiral polytopes having a group* $\mathrm{PSL}(3,q)$ *as automorphism group*, fixed 25-page author edition, 2022, Theorem 1.2 and Section 4.

[2] E. Schulte and A. I. Weiss, *Chiral polytopes*, DIMACS 4 (1991), 493–516; rotation correspondence and recursive intersection criterion, retained as in source Section 2.